Elham Esmaeilnia

SENIOR AI ENGINEER | LLM APPLICATIONS, INTELLIGENT AGENTS & AI SYSTEMS

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SUMMARY

AI Engineer building practical LLM applications and scalable AI systems, from RAG and intelligent agents to Persian NLP and speech. Hands-on with Python, PyTorch, FastAPI, and Docker, with experience in semantic retrieval, asynchronous services, and data-intensive pipelines. Combines applied R&D with end-to-end engineering to turn AI capabilities into reliable, deployable products.

PROFESSIONAL EXPERIENCE

Freelance AI Engineer | Independent Projects

Jan 2026 – Present

- Developing an LLM-driven support-triaging engine for automated ticket classification, semantic retrieval, and resilient processing.
- Building an asynchronous AI voice-sales-agent framework designed for low-latency interactions and reliable deployment.

Senior AI Engineer | Parnian Gostar

Aug 2025 – Jan 2026

- Designed and developed AlphaFusionNet, an AI-based financial modeling system.
- Implemented LLM-based intelligent agents for financial workflows; developed time-series pipelines and deep-learning models.
- Contributed to AI-driven hybrid decision-making systems.

AI Engineer | Bertina

Dec 2024 – Dec 2025

- Researched and developed Persian NLP systems, including text normalization and tokenization for grapheme-to-phoneme (G2P) workflows.
- Co-authored a G2P article from the Bertix Artificial Intelligence Group; contributed to language-understanding systems and a Persian diacritization model.

AI R&D Engineer | Farajahan

Nov 2023 – Dec 2024

- Developed intelligent question-answering and text-correction systems, including a RAG system grounded in Persian-language content.
- Developed a Persian speech-enhancement model and trained transformer-based language models.
- Created a multi-task Persian intelligent assistant using LLMs and CrewAI.

Big Data & Web Developer | Kanda Idea

Jun 2020 – Aug 2022

- Processed large-scale data using Spark, Hadoop, and HDFS; implemented analytics pipelines with Cassandra.
- Developed backend services, APIs, and web systems for IMIDRO and WEB MARINE; designed use-case and sequence diagrams.

AI Researcher | IPM

Apr 2018 – Aug 2018

- Conducted research on deep learning and semi-supervised processing.

Instructor | Alavi Institute

Jun 2013 – Jul 2014

- Taught programming, computer fundamentals, and mathematics.

SELECTED PROJECTS

AI-Based Financial Portfolio Optimization System
AI-based financial modeling and portfolio optimization developed in an industry setting. GitHub: github.com/Novoxpert/AlphaFusionNet

Persian NLP Pipeline: G2P, Tokenization & Text Correction
Persian-language NLP covering grapheme-to-phoneme processing, text normalization, tokenization, and text correction.

Persian Diacritization Model
Model designed to add diacritics to unmarked Persian text.

RAG-Based Intelligent Assistant
Question-answering system that generates responses grounded in Persian-language content using retrieval-augmented generation.

Persian Speech Enhancement & Multi-Task Assistant
Deep-learning speech enhancement and a Persian-language assistant using LLMs and CrewAI.

Voice Sales Agent
Scalable AI voice-agent framework for low-latency sales interactions, built with asynchronous patterns and modular architecture. GitHub: github.com/Elham-Esmaeilnia/voice-sales-agent

Web-Based Systems: IMIDRO & WEB MARINE
Web application and backend development delivered during the Big Data & Web Developer role.

TECHNICAL SKILLS

Programming & Backend: Python, C++, MATLAB, JavaScript, HTML/CSS, FastAPI, REST APIs, WebSockets

AI / ML / Speech: PyTorch, TensorFlow, Keras, Transformers, scikit-learn, OpenCV, Librosa, speech processing, deep learning, time-series modeling

LLMs & Agentic Systems: LLM applications, prompt engineering, RAG, LangChain, LangGraph, CrewAI, semantic retrieval, vector databases, structured outputs, tool calling

Data & Infrastructure: Spark, Hadoop, HDFS, Cassandra, MongoDB, Neo4j, Redis, Qdrant, FAISS, Docker, Docker Compose, Celery, RabbitMQ, Git

Automation & AI Development: n8n, MCP servers, Claude Code, Browser Use, LM Studio

Generative Media: Stable Diffusion, Adobe Firefly, Flux; Runway, Kling, Luma AI, Pika

EDUCATION

- PhD Candidate, Computer Engineering (AI & Robotics) — Islamic Azad University, Tehran Central
- MSc, Computer Engineering (Artificial Intelligence) — Rouzbehan University, Sari
- BSc, Computer Engineering — Shahid Beheshti University, Tehran

CERTIFICATIONS, HONORS & LANGUAGES

Certifications: Data Science & Python — DigiNext; Web Design — FanAsa; Teaching Skills — Ministry of Education; IT Internship — Agricultural Jihad Organization.

Honors: Selected Participant, National Mathematics Olympiad; City-Level Mathematics Olympiad; City-Level Physics Olympiad.

Languages: Persian (Native); English (Professional, EPT Score: 800).